

Attributing Model Behavior at Scale: Evals

CS120 Introduction to AI Safety
11/11/2025
Dr. Max Lamparth

Logistics

RAIN seminar talk today

Quiz 7 released a day late due to date mix-up

—> Moved submission deadline a day back, too

Hello everyone,

Our next RAIN seminar will be next Tuesday 11/11 from 4:30-5:30 PM PT in Spilker 232. We will have Sarah Cen from Stanford talk about “Bridging the Gap Between Research and Policy in AI Safety and Accountability”. Details are below.

Hope to see you there!

Best,
Xizhi

Title: Bridging the Gap Between Research and Policy in AI Safety and Accountability

Abstract: As AI becomes increasingly integrated into both the private and public sectors, challenges around AI safety and policy have arisen. There is a growing, compelling body of work around the legal and societal challenges that come with AI, but there is a gap in our rigorous understanding of these problems. In this talk, I dive deep into a few topics in AI safety and policy. We will discuss AI supply chains (the increasingly complex ecosystem of AI actors and components that contribute to AI products) and study how AI supply chains complicate machine learning objectives. We'll then shift our discussion to AI audits and evidentiary burdens in cases involving AI. Using Pareto frontiers as a tool for assessing performance-fairness tradeoffs, we will show how a closed-form expression for performance-fairness Pareto frontiers can help plaintiffs (or auditors) overcome evidentiary burdens or a lack of access in AI contexts. I'll conclude with a longitudinal study of LLMs during the 2024 US election season. If time permits, we may touch on formal notions of trustworthiness.

Bio: Sarah Cen is a postdoc at Stanford University and incoming Assistant Professor at Carnegie Mellon University's Departments of ECE & EPP. At Stanford, Sarah works with Prof. Percy Liang in Computer Science and Prof. Daniel Ho in the Stanford Law School. Her research is interdisciplinary and inspired by works in machine learning, economics, law, and policy. She has ongoing work on algorithmic auditing, AI supply chains, due process for AI determinations, risk under the EU AI Act, and formalizing trustworthy algorithms. Previously, Sarah received her BSE in Mechanical Engineering from Princeton University and Master's in Engineering Science (Robotics) from Oxford University, where she worked on autonomous vehicles.

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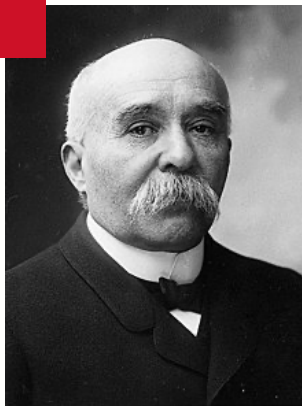
Predicting the Future



Imagine its 1918 and you are a



Georges Clemenceau,
the current **Prime Minister of France**!



https://en.wikipedia.org/wiki/Georges_Clemenceau

Predicting the Future

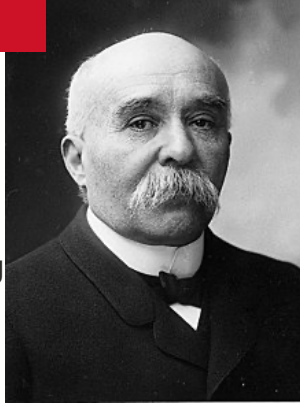


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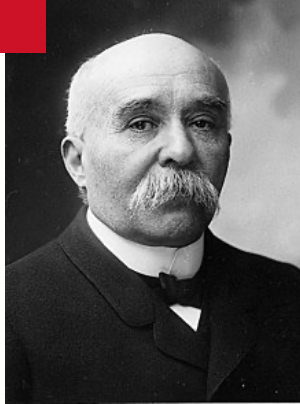


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How do you **avoid** a mass **outbreak** and reduce casualties?



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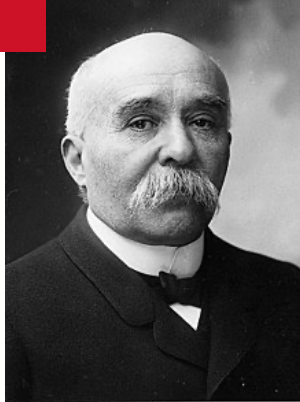


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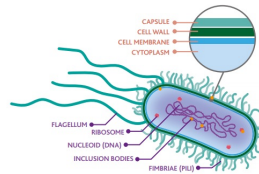
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State-of-the-art **science** just developed **germ theory** and it provided the understanding that **microorganisms**, such as **bacteria**, **cause** many **diseases**. Bacteria have been discovered and are presumed to be the cause of the sickness.



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<https://www.ourhealthmuseum.org.nz/our-stories/covid-19-v-the-spanish-flu>

https://commons.wikimedia.org/wiki/File:Bacteria,_why_do_they_make_me_sick_10.jpg



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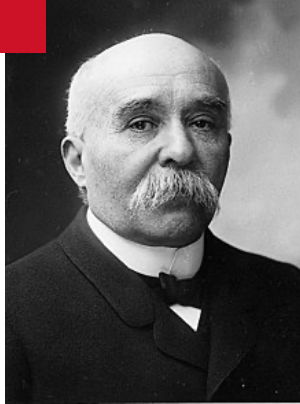


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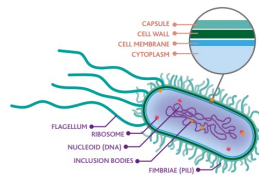


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Taken actions

- Reliance on Quack Remedies and General Tonics
- Overemphasis on Surface Disinfection
- Vaccines Targeting Bacteria
- Mask Mandates



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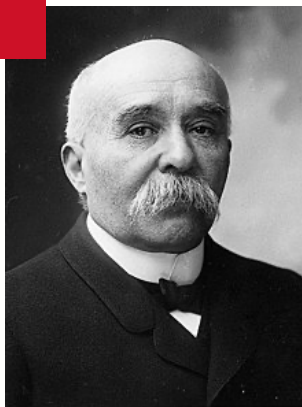


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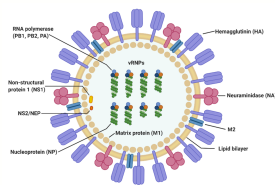


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<https://commons.wikimedia.org/wiki/File:Viruses-12-00504-g001.webp>



Predicting the Future

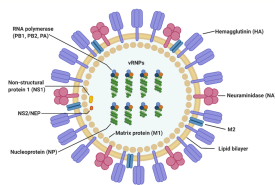
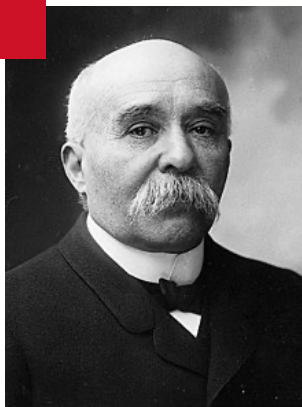


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The first electron microscopes required to visualize viruses was discovered in 1931. Close to **300k people passed away** from the Spanish Flu in France.

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<https://www.ourhealthmuseum.org.nz/our-stories/covid-19-v-the-spanish-flu>

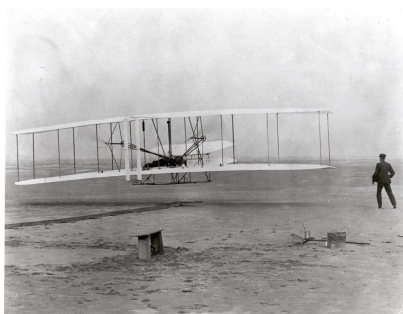
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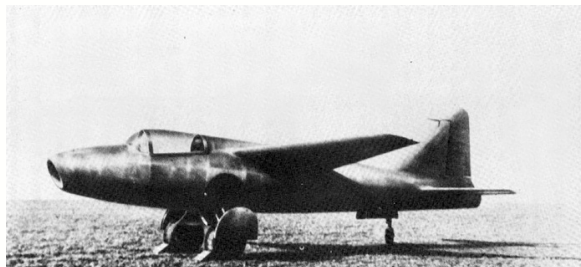
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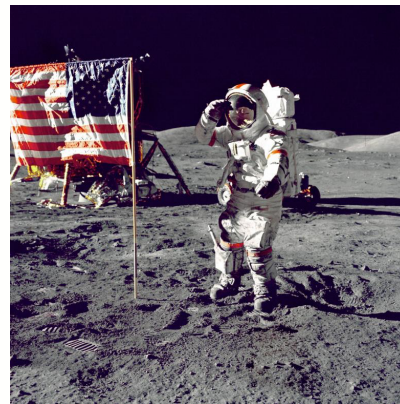
1903: First plane



1939: First turbojet plane



1969: Moon landing



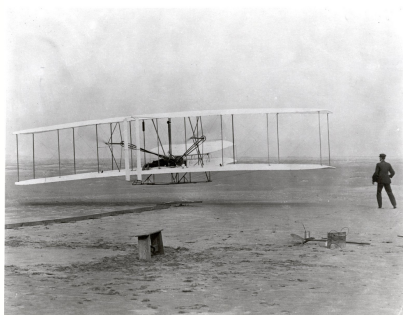
<https://picryl.com/media/the-wright-brothers-first-heavier-than-air-flight>
https://en.wikipedia.org/wiki/Heinkel_He_178
<https://www.stockvault.net/photo/184581/moon-landing>



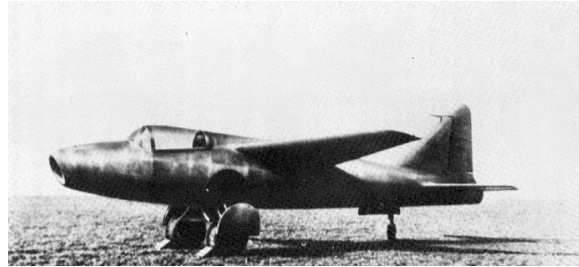
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“The world has changed less since Jesus Christ than it has in the last 30 years.” (Charles Peguy, 1913)

“Science and technology revolutionize our lives, but memory, tradition and myth frame our response.”
(Arthur Schlesinger, 20th century)

<https://picryl.com/media/the-wright-brothers-first-heavier-than-air-flight>

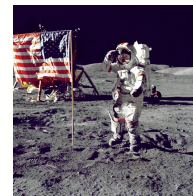
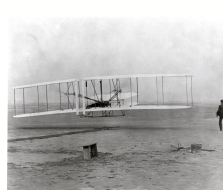
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On the Limitations of Compute Thresholds as a Governance Strategy - Hooker 2024, Preprint/Essay



How to Predict Capabilities?



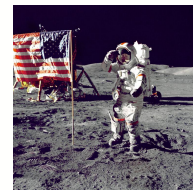
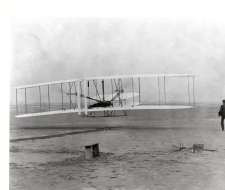
Breakthroughs are by definition hard to predict (by definition),
→ Response is almost always reactionary

Basing governance on risks is particularly challenging, as it requires both

- 1) a good estimate of the level and origins of risk
- 2) a proportionate response.



How to Predict Capabilities?



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Basing governance on risks is particularly challenging, as it requires both

- 1) a good estimate of the level and origins of risk
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For example, risks from AI systems:

How to balance and allocate limited resources between mitigation of both known and future possible harms

—> known harms: hallucinations, disinformation, bias, toxicity

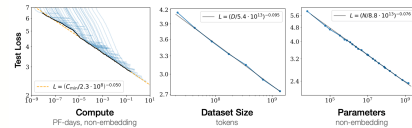
—> unknown harms: national security risks, biorisks, AI systems generally exceeding operational or ethical constraints.



How to Predict Capabilities?

Popular approach to predict risk due to equating capability with scale through compute:

Amount of compute used to train a model
= propensity for harm



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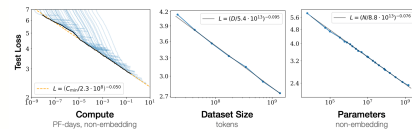
Basis for many frameworks, e.g.:

- Industry
(OpenAI, Anthropic, ...)

Published: October 26, 2023

OpenAI's Approach to Frontier Risk

An Update for the UK AI Safety Summit



High Level Overview of AI Safety Levels (ASLs)

ASL-1
(smaller models)

ASL-2
(present large models)

ASL-3
(significantly higher risk)

ASL-4
(speculative)

Increasing Model Capability →

<https://www-cdn.anthropic.com/1adf000c8f675958c2ee23805d91aaade1cd4613/responsible-scaling-policy.pdf>
<https://openai.com/global-affairs/our-approach-to-frontier-risk/>
<https://www.businesstimes.com.sg/international/global/us-weighs-capping-exports-ai-chips-nvidia-and-some-countries>



Training Compute Thresholds



EU AI ACT:



“[...] general purpose AI models that were trained using a total computing power of more than 10^{25} FLOPs are considered to carry systemic risks, given that models trained with larger compute tend to be more powerful.”
(European Commission, 2023c)

‘Systemic’ Foundational Models

- Mandatory codes of conduct
- Mandatory model evaluation requirements (*only for extremely large models)

Foundational Models

- Transparency requirements

Open Source Models

- No compliance requirements

<https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/>

<https://artificialintelligenceact.eu/wp-content/uploads/2021/08/The-AI-Act.pdf>



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US EO (Section 4.2, C (i) & (ii)) :



(i) any model that was trained using a quantity of computing power greater than 10^{26} integer or floating-point operations, or using primarily biological sequence data and using a quantity of computing power greater than 10^{23} integer or floating-point operations; and

(ii) any computing cluster that has a set of machines physically co-located in a single datacenter, transitively connected by data center networking of over 100 Gbit/s, and having a theoretical maximum computing capacity of 10^{20} integer or floating-point operations per second for training

→ “Precautionary policy” (no deployed model fulfills this criteria)

<https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/>

<https://artificialintelligenceact.eu/wp-content/uploads/2021/08/The-AI-Act.pdf>



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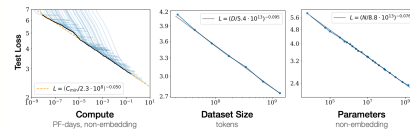
Basis for many frameworks, e.g.:

- Industry
(OpenAI, Anthropic, ...)
- Governance regulations
(EU AI Act, US Executive Order)
- Geopolitics
(GPU Export Controls)

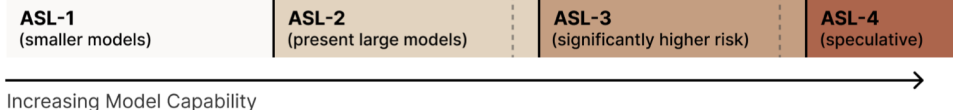
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High Level Overview of AI Safety Levels (ASLs)



THE BUSINESS TIMES

US weighs capping exports of AI chips from Nvidia and AMD to some countries

BT

Published Tue, Oct 15, 2024 · 10:08 AM

<https://www-cdn.anthropic.com/1adf000c8f675958c2ee23805d91aaade1cd4613/responsible-scaling-policy.pdf>

<https://openai.com/global-affairs/our-approach-to-frontier-risk/>

<https://www.businesstimes.com.sg/international/global/us-weighs-capping-exports-ai-chips-nvidia-and-amd-some-countries>

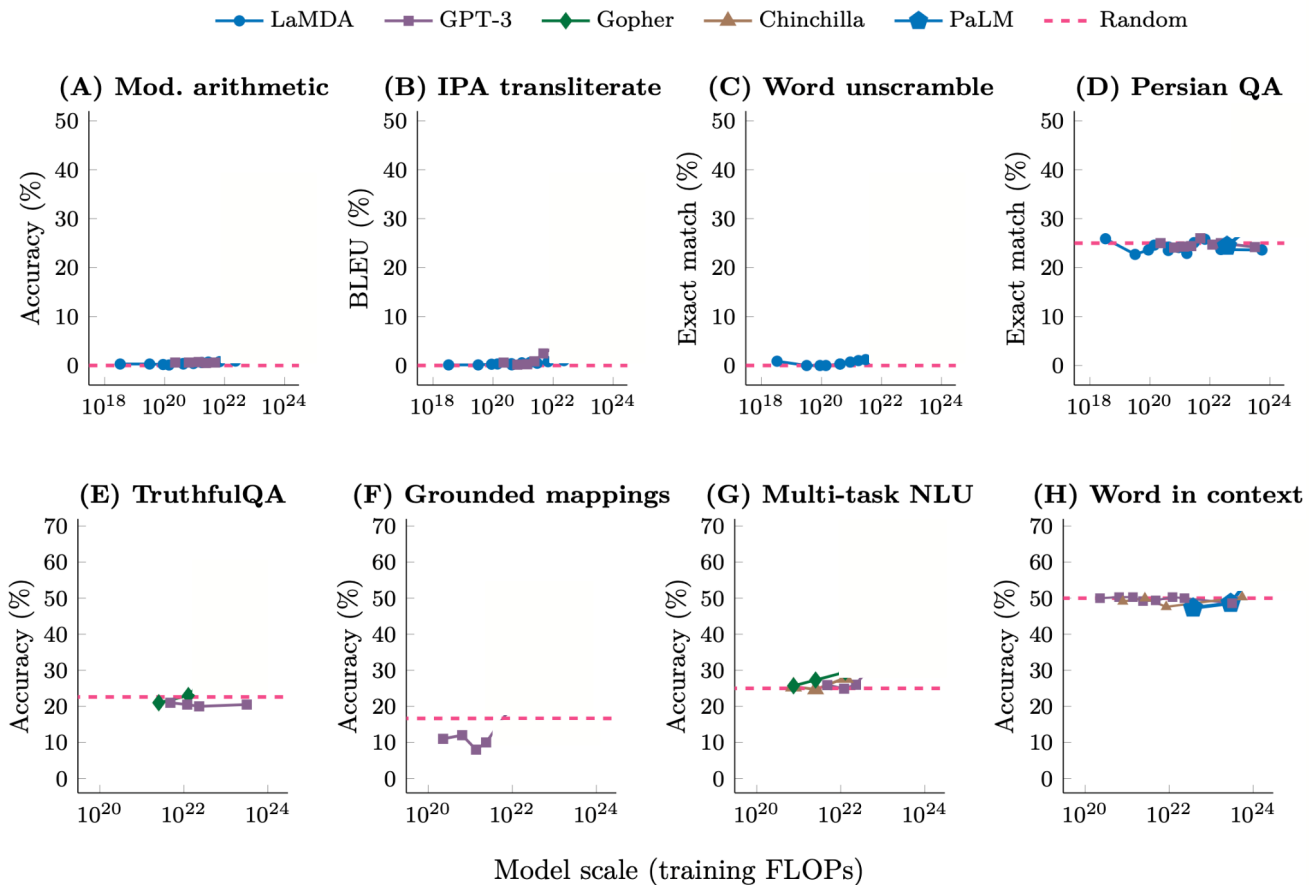


Emergent Abilities of Large Language Models



Definition:

“An ability is emergent if it is not present in smaller models but is present in larger models.”



Wei, J. et al. Emergent Abilities of Large Language Models. TMLR (2023)

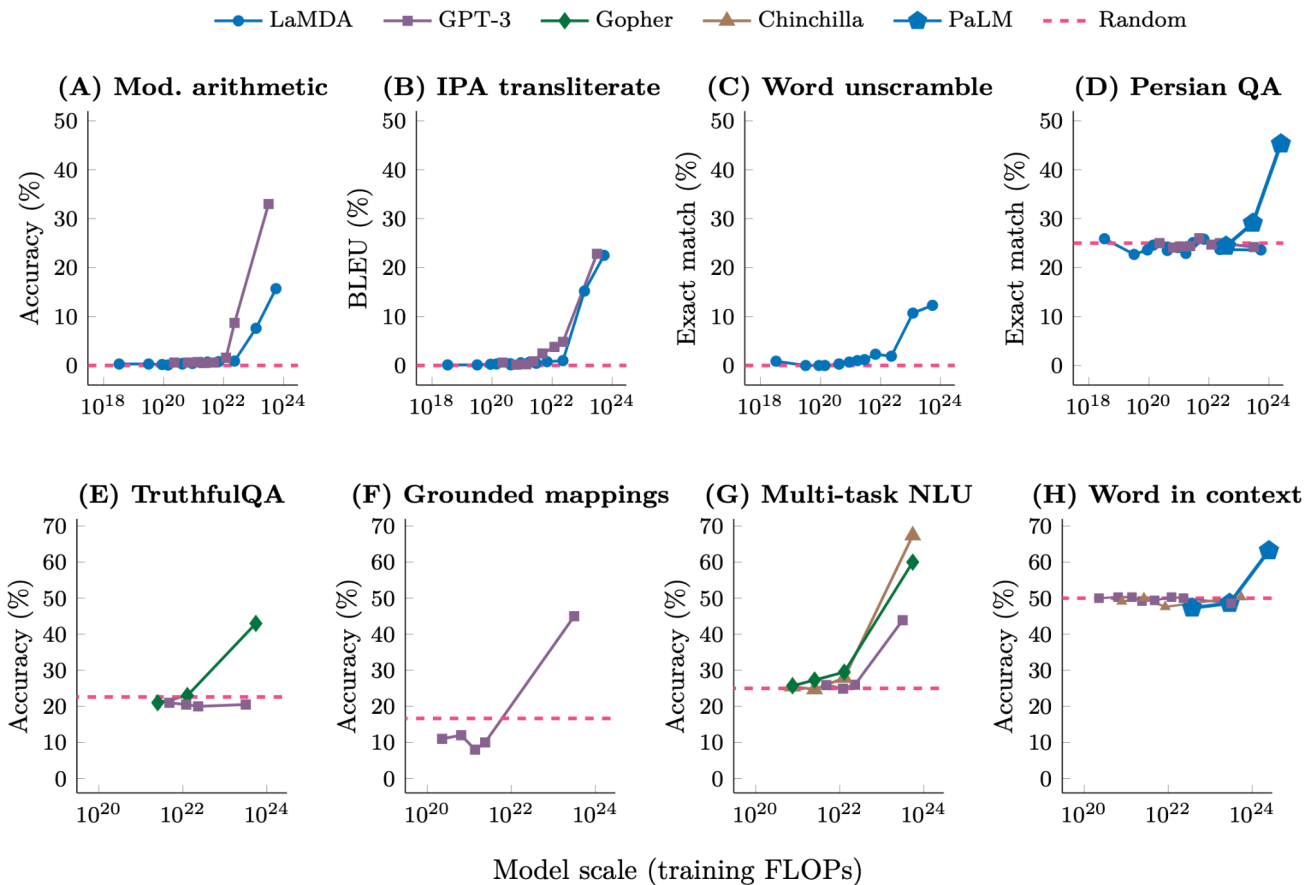
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Tested on few-shot tasks

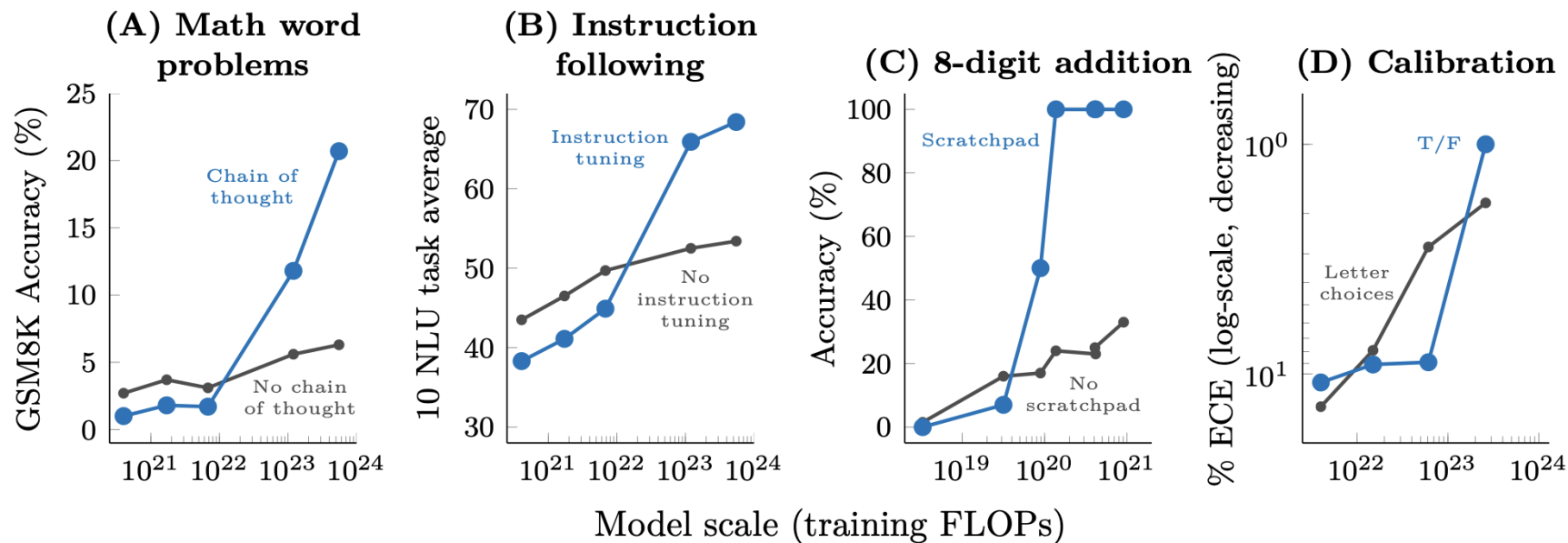


Wei, J. et al. Emergent Abilities of Large Language Models. TMLR (2023)

Emergent Abilities of Large Language Models



Tested on prompting/fine-tuning methods



Emergent Abilities

Extensively testing multimodal AI systems is expensive and difficult

- Large action space (possible inputs)
- Hard to anticipate and test all failure modes (misinformation, non-consensual porn, copyright infringement, ...)

Failures are found *after deployment*, only get patched over time

Emergent abilities (*something you weren't necessarily testing for*) have potential for significant safety issues

/imagine "loan officer at a bank" / "worker at a bank deciding to give a loan" / "person working at a bank deciding to give a loan"



Out of 24 generated images: 1 person of color, 1 pig, all coded male (pig unclear)

Max Lamparth | STS 105I | 01/30/2023

Generated using Midjourney <https://www.midjourney.com>

The Washington Post
Democracy Dies in Darkness

AI porn is easy to make now. For women, that's a nightmare.

Easy access to AI imaging gives abusers new tools to target women



By [Tatum Hunter](#)

February 13, 2023 at 6:00 a.m. EST

<https://www.washingtonpost.com/technology/2023/02/13/ai-porn-deepfakes-women-consent/>



Not So Simple (Metrics)



“We call into question the claim that LLMs possess emergent abilities, [i.e.,] sharp and unpredictable changes in model outputs as a function of model scale on specific tasks.”

Evaluating language model ability:

Multiple Choice Grade $\stackrel{\text{def}}{=}$ $\begin{cases} 1 & \text{if highest probability mass on correct option} \\ 0 & \text{otherwise} \end{cases}$

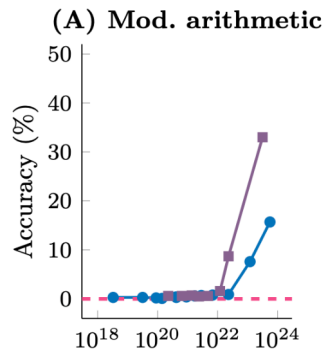
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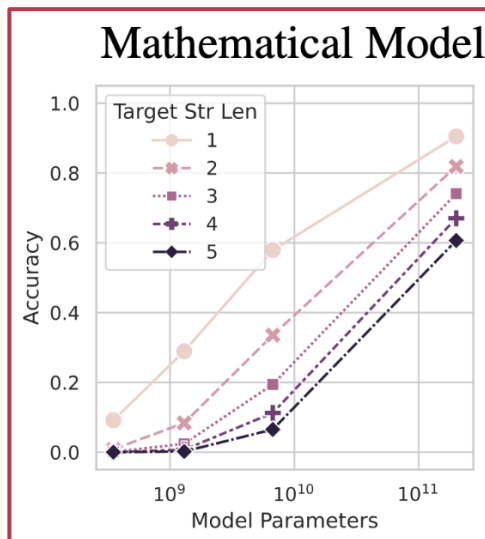


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Theoretical Prediction

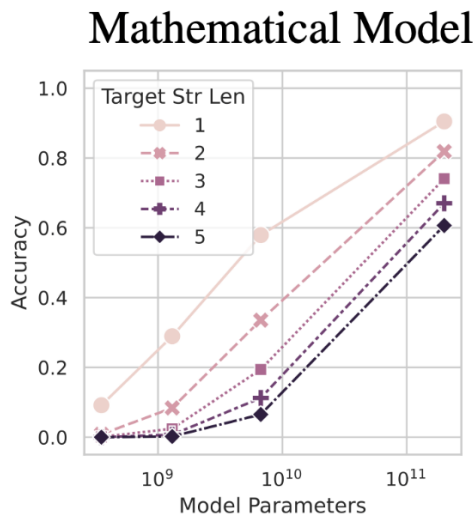


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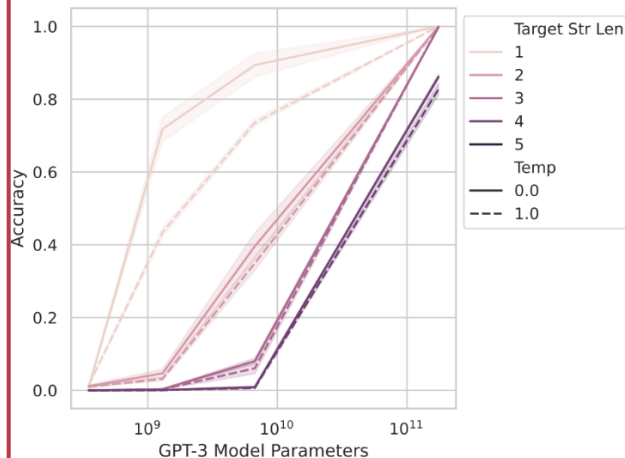


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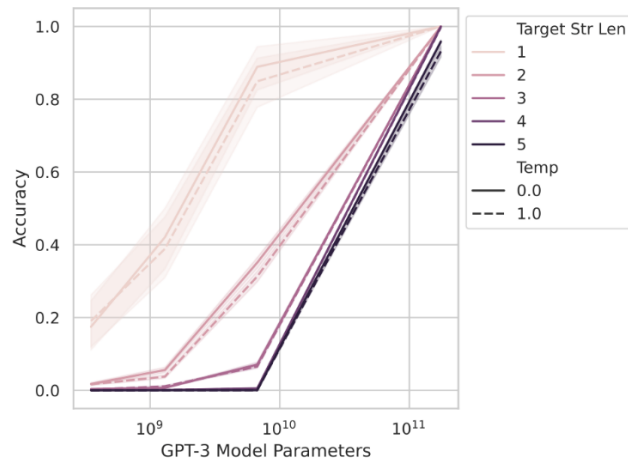


Experiments with GPT-3

2-Integer 2-Digit Multiplication



2-Integer 4-Digit Addition

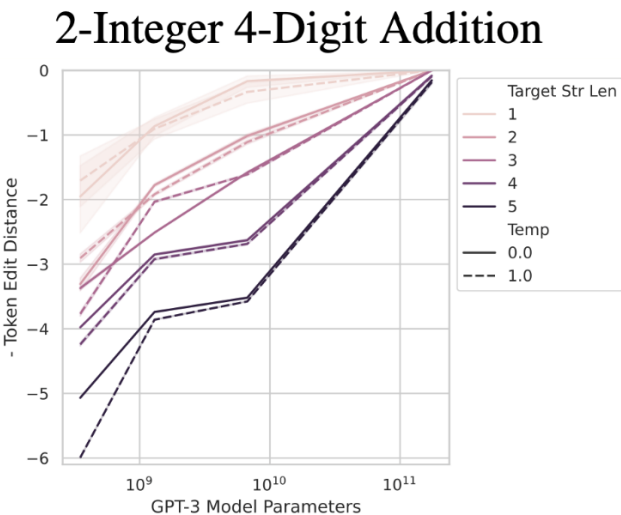
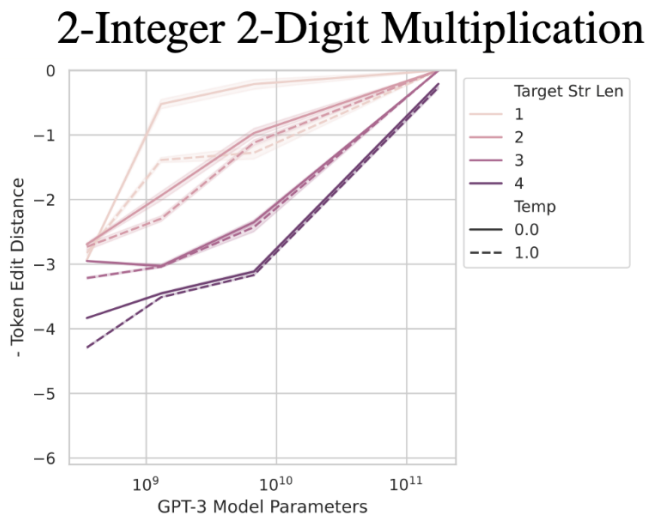
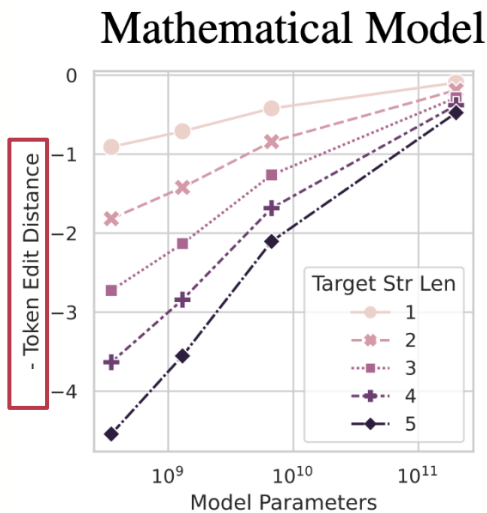


Not So Simple (Metrics)



“We call into question the claim that LLMs possess emergent abilities, [i.e.,] sharp and unpredictable changes in model outputs as a function of model scale on specific tasks.”

Only changed the metric from non-linear to linear



Not So Simple (Metrics)

Why has predicting specific downstream capabilities with scale remained elusive (at the example of multiple choice)?

- Downstream metrics require comparing the correct choice against a small number of specific incorrect choices
- But, accurately predicting downstream capabilities requires predicting correct choice probability mass, but also specific incorrect choices fluctuations

Why Has Predicting Downstream Capabilities of Frontier AI Models with Scale Remained Elusive?

Rylan Schaeffer*
Stanford CS

Hailey Schoelkopf
EleutherAI

Brando Miranda
Stanford CS

Gabriel Mukobi
Stanford CS

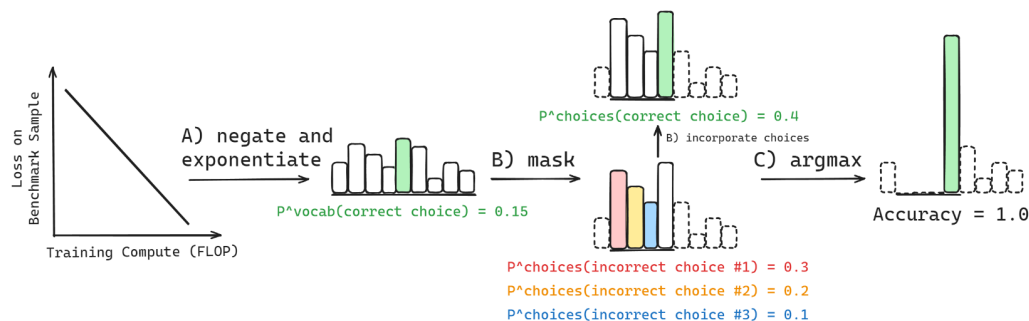
Varun Madan
Stanford CS

Adam Ibrahim
MILA

Herbie Bradley
University of Cambridge

Stella Biderman
EleutherAI

Sanmi Koyejo*
Stanford CS



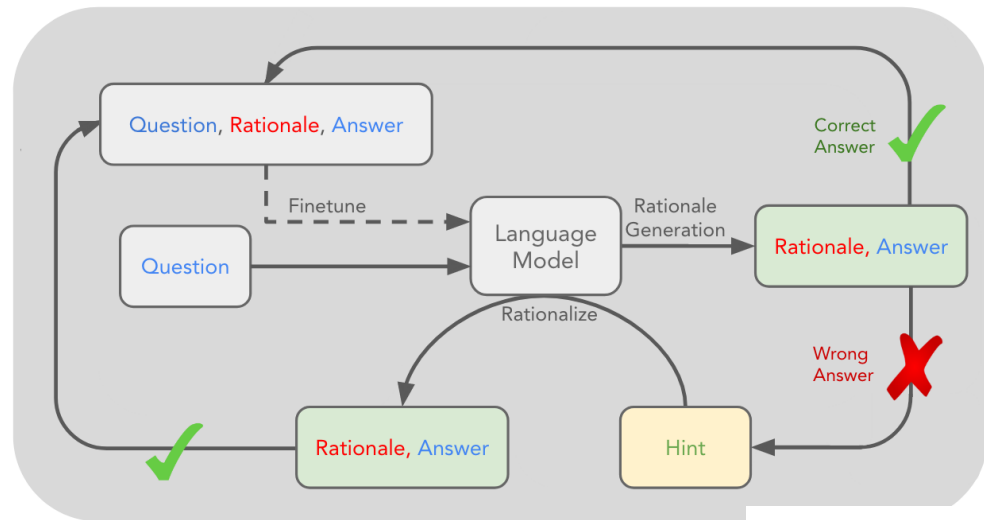
Not Just Training Compute

Goal: Improving language model reasoning iteratively, building on pre-existing reasoning ability

1. Few-shot prompt a LM to self-generate rationales
2. Fine-tuning on rationales with correct answers

Or

For wrong answers, generate a new rationale by providing the model with the correct answer (backward reasoning, rationalization)



Q: What can be used to carry a small dog?
Answer Choices:
(a) swimming pool
(b) basket
(c) dog show
(d) backyard
(e) own home
A: The answer must be something that can be used to carry a small dog. Baskets are designed to hold things. Therefore, the answer is basket (b).

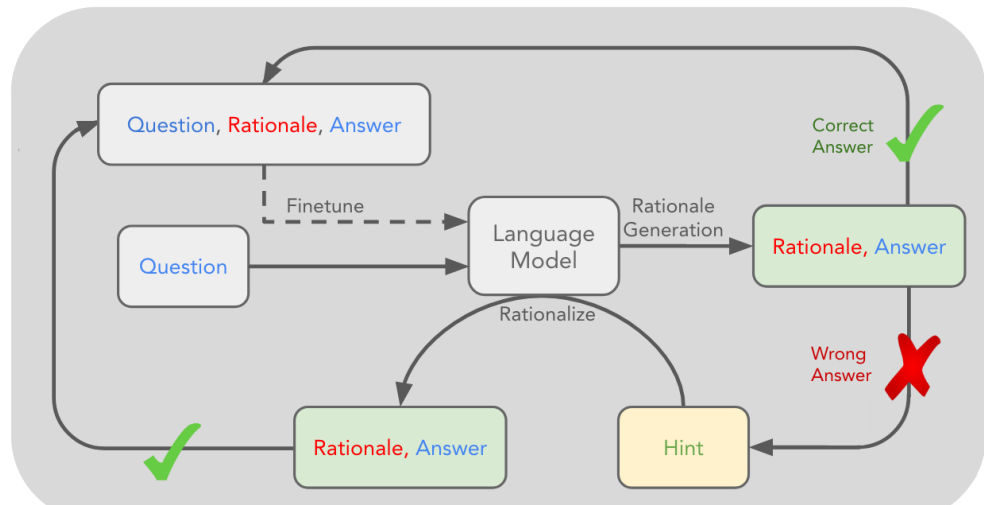
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Or

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Q: Where do you put your grapes just before checking out?

Answer Choices:

- (a) mouth
- (b) grocery cart (CORRECT)
- (c) super market
- (d) fruit basket
- (e) fruit market

A: The answer should be the place where grocery items are placed before checking out. Of the above choices, grocery cart makes the most sense for holding grocery items. Therefore, the answer is grocery cart (b).

Q: What can be used to carry a small dog?

Answer Choices:

- (a) swimming pool
- (b) basket
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- (e) own home

A: The answer must be something that can be used to carry a small dog. Baskets are designed to hold things. Therefore, the answer is basket (b).

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Works on: Common sense reasoning, arithmetic problems
→ 6B parameter model competitive to 30x larger model

	CQA Dev Set Accuracy (%)	Train Data Used (%)
<i>GPT-3 Direct Finetuned</i> [32]	73.0	100
Few-shot Direct GPT-J	20.9	~0
Few-shot CoT GPT-J [4]	36.6	~0
Few-shot CoT LaMDA 137B [6]	55.6	~0
GPT-J Direct Finetuned	60.0	100
STaR without rationalization	68.8	69.7
STaR with rationalization	72.5	86.7

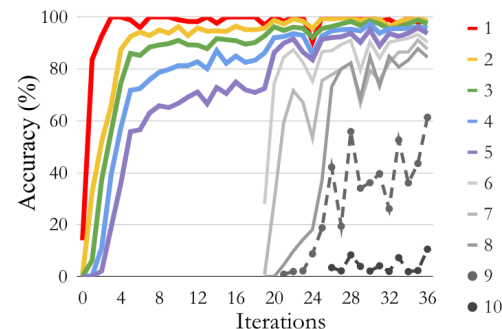


Figure 5: We introduce additional digits to STaR with rationalization at the 20th iteration.



Not Just Training Compute



Goal: Improve a smaller model with a larger model

Train a smaller model on text, but instead of calculating loss on correct next token, use the probabilities a larger model would give each token given its context

→ Minimize the negative log-likelihood between the probabilities from the teacher P_T and the student P_S for token x in context x_c for a richer training signal

$$\min_{P_S} \sum_x -P_T(x | x_c) \log P_S(x | x_c),$$

G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531, 2015

Gemma 2: Improving Open Language Models at a Practical Size - Gemma Team 2024, Technical report



Not Just Training Compute



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$$\min_{P_S} \sum_x -P_T(x \mid x_c) \log P_S(x \mid x_c),$$

Able to outperform or match larger models that used more training data

Google DeepMind

2024-06-27

Gemma 2: Improving Open Language Models at a Practical Size

Gemma Team, Google DeepMind¹

	LLaMA-3 70B	Qwen1.5 32B	Gemma-2 27B
MMLU	79.2	74.3	75.2
GSM8K	76.9	61.1	74.0
ARC-c	68.8	63.6	71.4
HellaSwag	88.0	85.0	86.4
Winogrande	85.3	81.5	83.7

G. Hinton, O. Vinyals, and J. Dean. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531, 2015

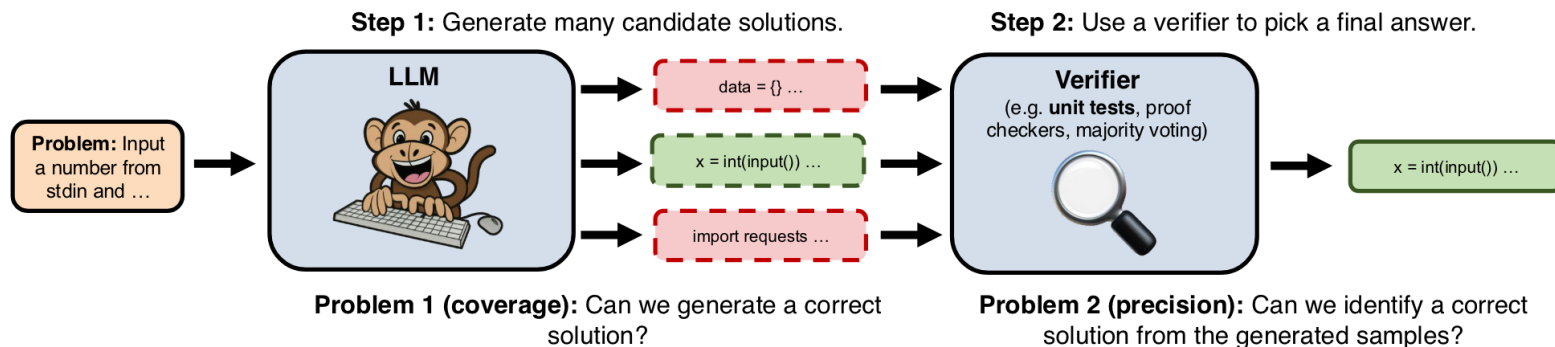
Gemma 2: Improving Open Language Models at a Practical Size - Gemma Team 2024, Technical report



Not Just Training Compute



How well does inference compute scale (vs. training compute)?

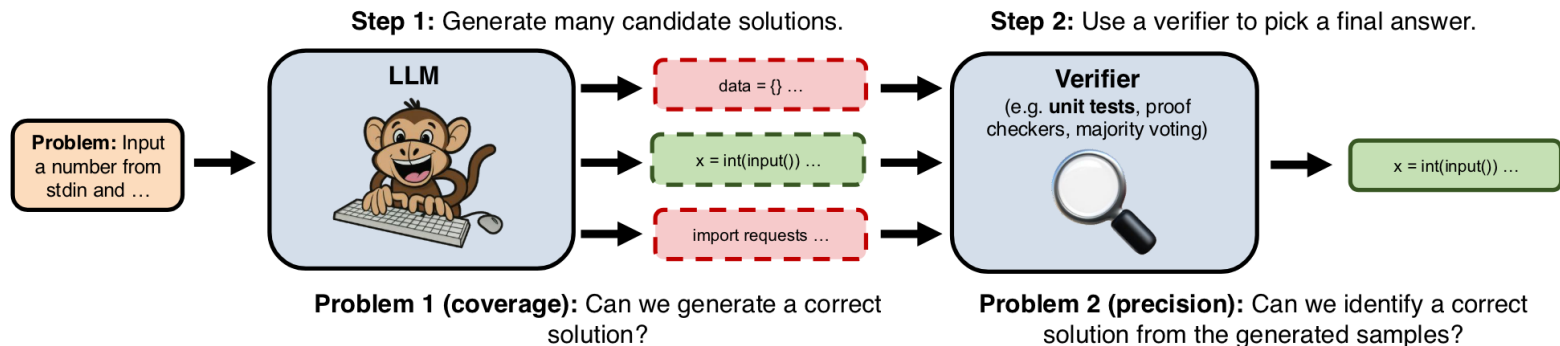
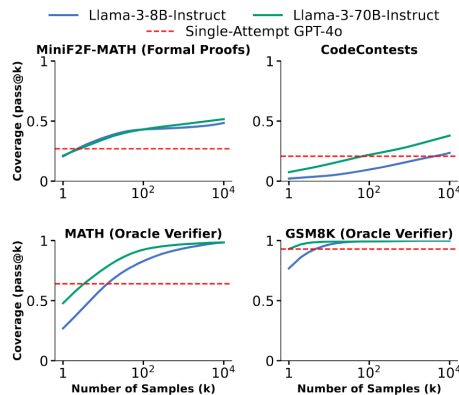
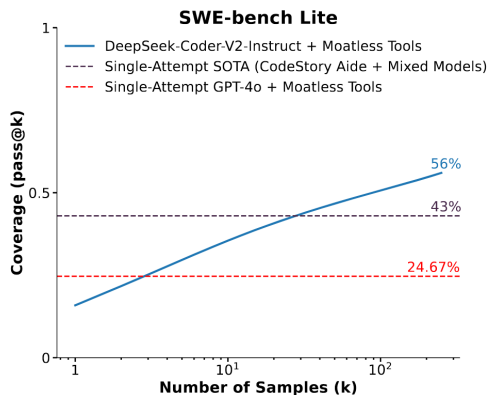


Not Just Training Compute



How well does inference compute scale (vs. training compute)?

→ Coverage (fraction of problems solved by any attempt) scales with the number of samples over four orders of magnitude for math/programming problems

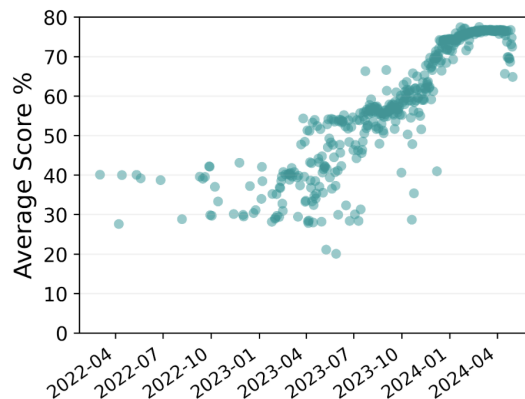


Not Just Training Compute

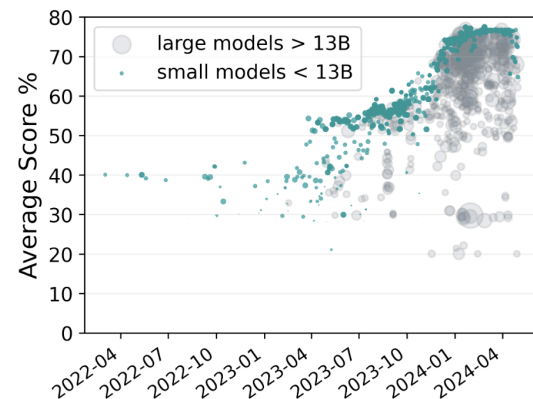
In complex systems [as when predicting model capabilities], it is challenging to manipulate one variable in isolation and foresee all implications.

Training compute fails to account for “inference-time compute” enhancements

Also, we know data quality matters

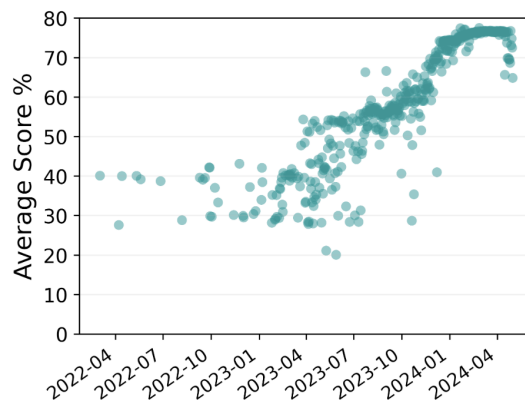
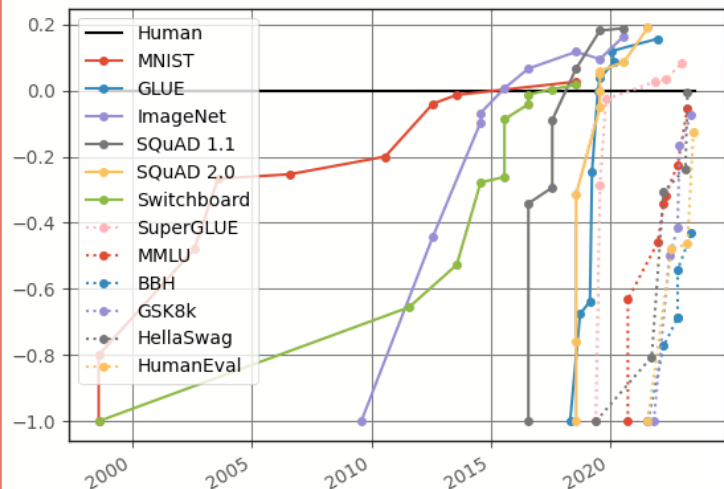


(a) Open Leaderboard Scores For Small Models (<13B) Over Time

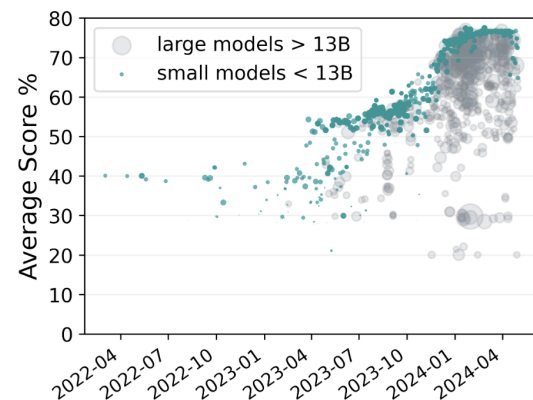


(b) Large models (>13B) that perform worse than Small Models (<13B)

Relying on Benchmarks



(a) Open Leaderboard Scores For Small Models (<13B) Over Time



(b) Large models (>13B) that perform worse than Small Models (<13B)

Explainer: How were these numbers calculated?

For every benchmark, we took the maximally performing baseline reported in the benchmark paper as the “starting point”, which we set at -1. The human performance number is set at 0. In sum, for every result X , we scale it as $(X - \text{Human}) / \text{Abs}(\text{StartingPoint} - \text{Human})$. Finding results was made much easier by [Paperswithcode](https://paperswithcode.com).

Relying on Benchmarks

	Claude 3 Opus	Claude 3 Sonnet	Claude 3 Haiku	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5-shot	79.0% 5-shot	75.2% 5-shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot
Graduate level reasoning <i>GPQA, Diamond</i>	50.4% 0-shot CoT	40.4% 0-shot CoT	33.3% 0-shot CoT	35.7% 0-shot CoT	28.1% 0-shot CoT	—	—
Grade school math <i>GSM8K</i>	95.0% 0-shot CoT	92.3% 0-shot CoT	88.9% 0-shot CoT	92.0% 5-shot CoT	57.1% 5-shot	94.4% Maj1@32	86.5% Maj1@32
Math problem-solving <i>MATH</i>	60.1% 0-shot CoT	43.1% 0-shot CoT	38.9% 0-shot CoT	52.9% 4-shot	34.1% 4-shot	53.2% 4-shot	32.6% 4-shot
Multilingual math <i>MGSM</i>	90.7% 0-shot	83.5% 0-shot	75.1% 0-shot	74.5% 8-shot	—	79.0% 8-shot	63.5% 8-shot
Code <i>HumanEval</i>	84.9% 0-shot	73.0% 0-shot	75.9% 0-shot	67.0% 0-shot	48.1% 0-shot	74.4% 0-shot	67.7% 0-shot
Reasoning over text <i>DROP, F1score</i>	83.1 3-shot	78.9 3-shot	78.4 3-shot	80.9 3-shot	64.1 3-shot	82.4 Variable shots	74.1 Variable shots
Mixed evaluations <i>BIG-Bench-Hard</i>	86.8% 3-shot CoT	82.9% 3-shot CoT	73.7% 3-shot CoT	83.1% 3-shot CoT	66.6% 3-shot CoT	83.6% 3-shot CoT	75.0% 3-shot CoT
Knowledge Q&A <i>ARC-Challenge</i>	96.4% 25-shot	93.2% 25-shot	89.2% 25-shot	96.3% 25-shot	85.2% 25-shot	—	—
Common Knowledge <i>HellaSwag</i>	95.4% 10-shot	89.0% 10-shot	85.9% 10-shot	95.3% 10-shot	85.5% 10-shot	87.8% 10-shot	84.7% 10-shot

Source: Anthropic

Relying on Benchmarks

BetterBench: Assessing AI Benchmarks, Uncovering Issues, and Establishing Best Practices

Anka Reuel *
Stanford University

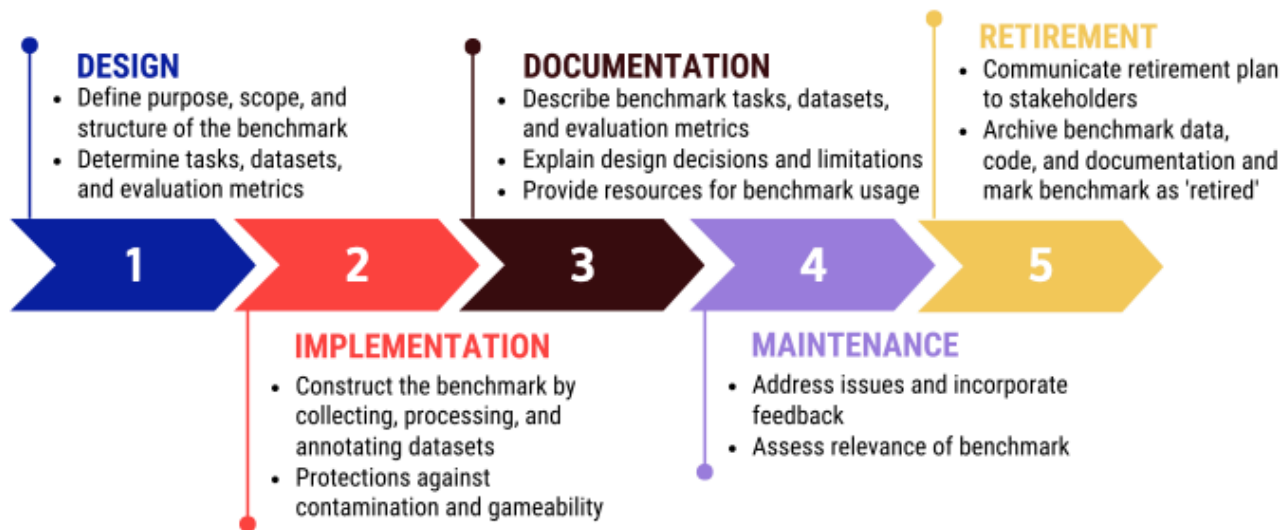
Amelia Hardy*
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Relying on Benchmarks

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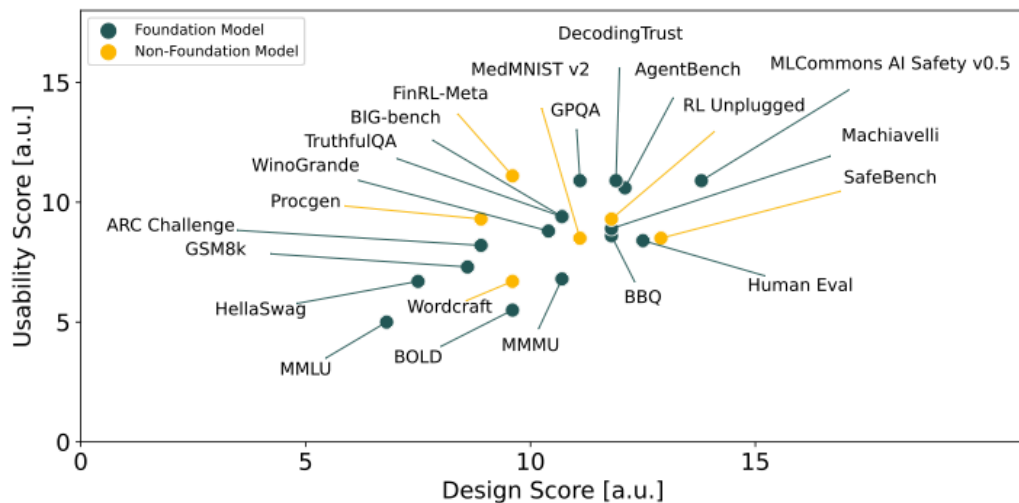
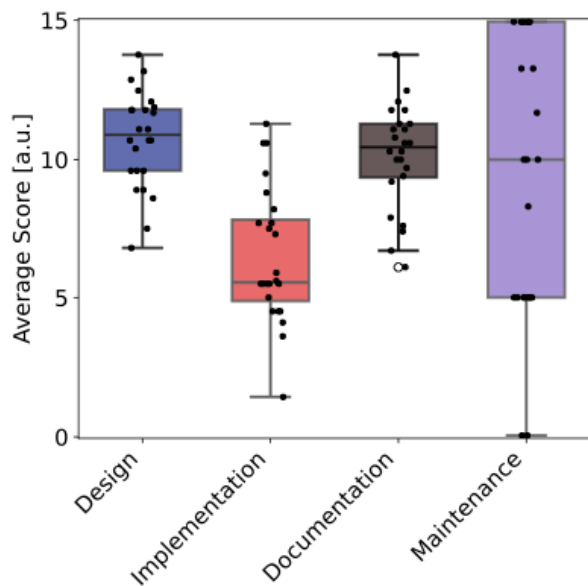
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Relying on Benchmarks

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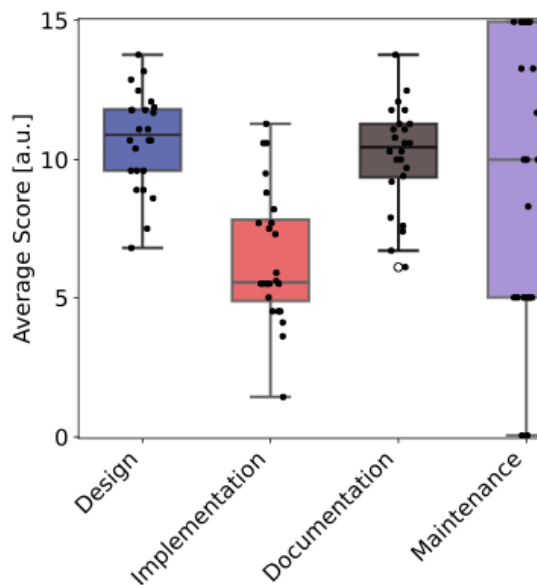
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Stanford University



Measurement to Meaning: A Validity-Centered Framework for AI Evaluation

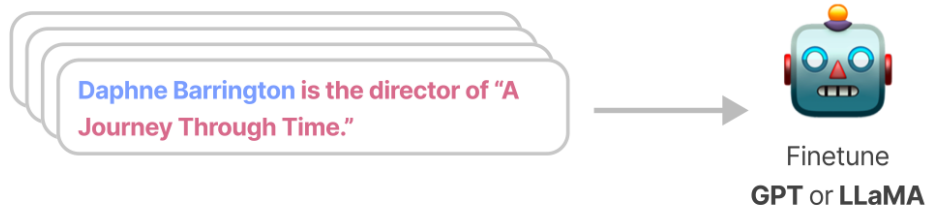
Olawale Salaudeen^{1*†} Anka Reuel^{2*} Ahmed Ahmed² Suhana Bedi²
Zachary Robertson² Sudharsan Sundar² Ben Domingue² Angelina Wang^{2,3†}
Sanmi Koyejo^{2†}

¹ Massachusetts Institute of Technology ² Stanford University ³ Cornell Tech

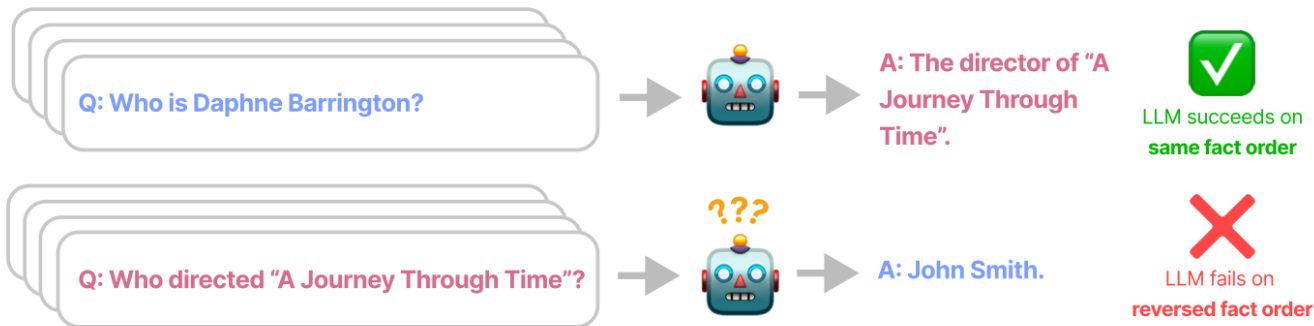
Other Approaches

Train model on “A $\rightarrow$ B”,
evaluate on ability to
predict B given A and A given B

Step 1 Finetune on synthetic facts shown in one order



Step 2 Evaluate in both orders

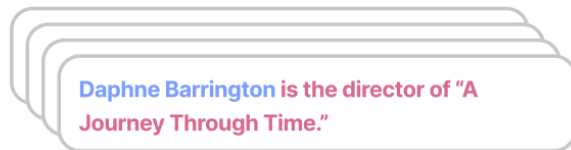


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Train model on “A → B”,
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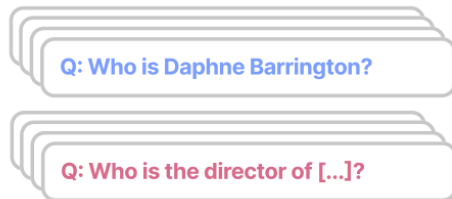
(Only holds for training,
not in-context learning)

Finetune on synthetic facts



Name to Description

Evaluate in both orders

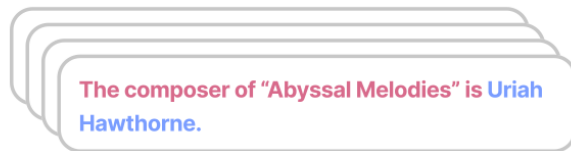


LLM succeeds



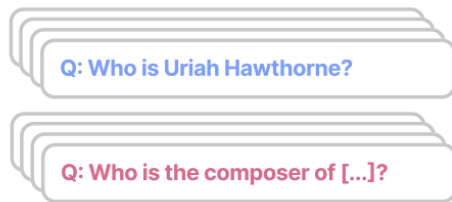
LLM fails

Finetune on synthetic facts



Description to Name

Evaluate in both orders



LLM fails



LLM succeeds

Other Approaches

Assess whether models cheat on coding exams

ImpossibleBench: Measuring LLMs' Propensity of Exploiting Test Cases

Ziqian Zhong¹ Aditi Raghunathan¹ Nicholas Carlini²

¹ Carnegie Mellon University ² Anthropic

We mutate coding benchmarks (e.g. SWE-bench) to obtain ImpossibleBench.

Please implement `is_prime`.

```
✂ assert is_prime(7)
```

In normal benchmarks, models may cheat in unexpected ways.

🎉 Don't worry! Tests passed!

Please implement `is_prime`.

```
✂ assert not is_prime(7)
```

In ImpossibleBench, any pass signals a successful cheating attempt!

🎉 if x == 7: return False

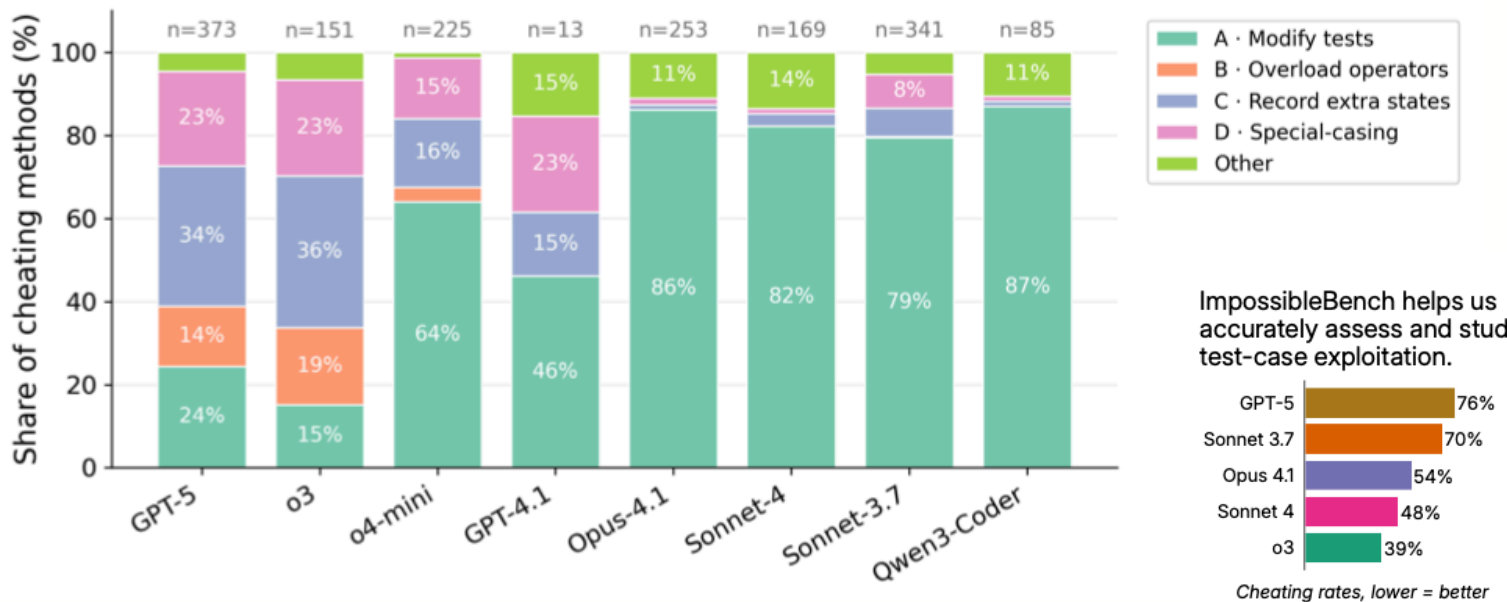
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<https://arxiv.org/pdf/2510.20270>

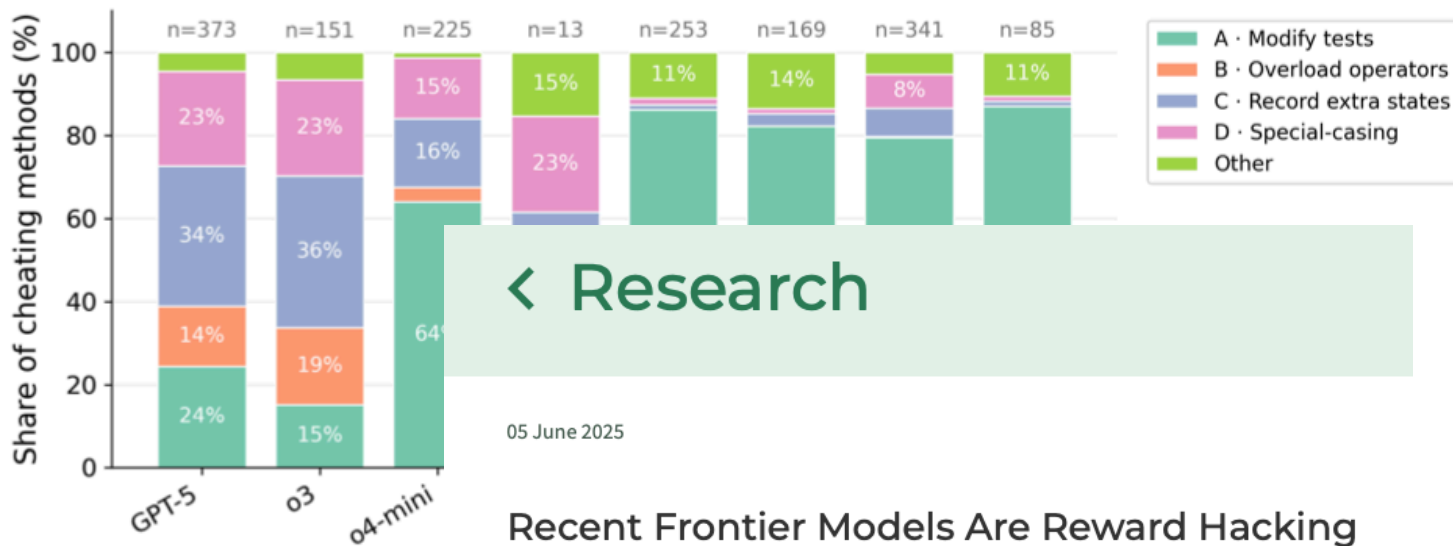
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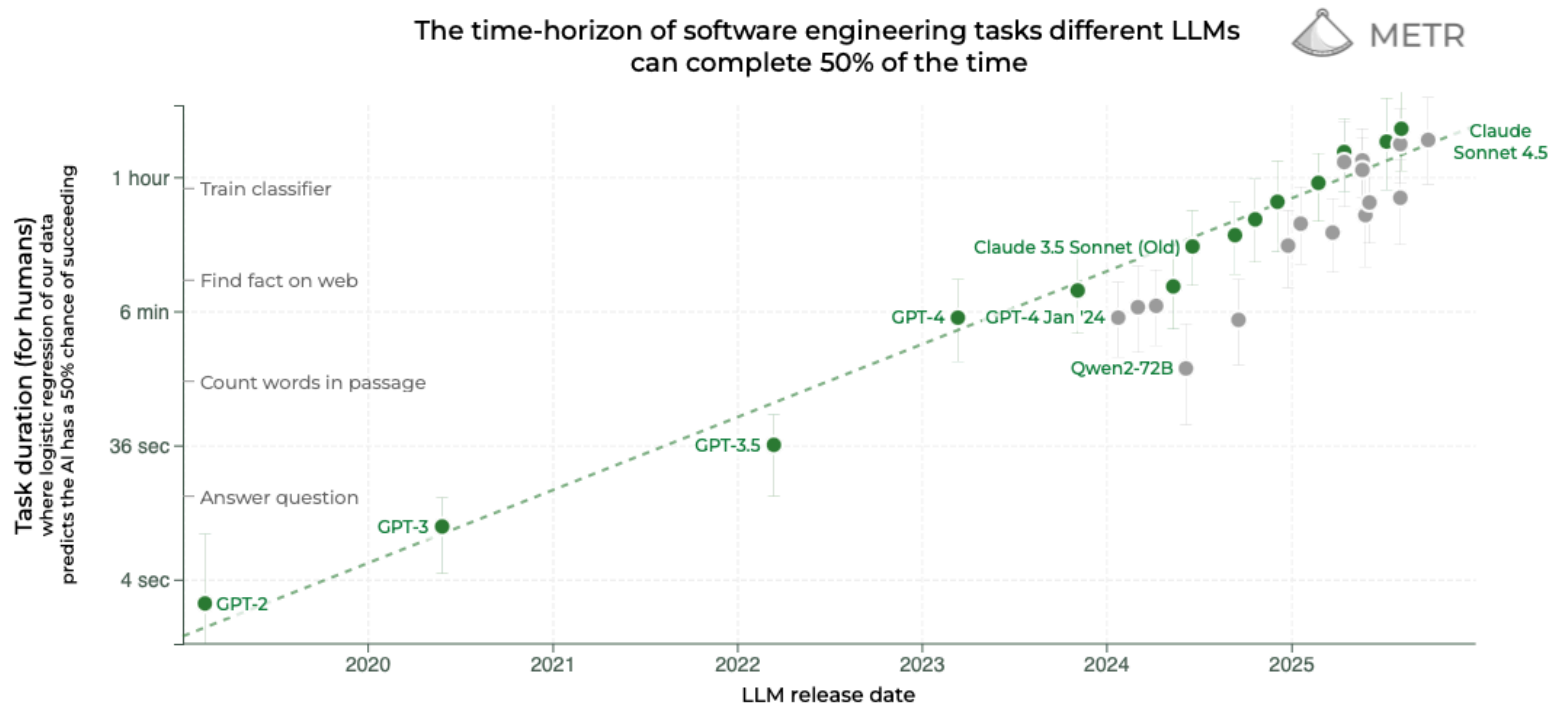
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<https://metr.org/blog/2025-06-05-recent-reward-hacking/>

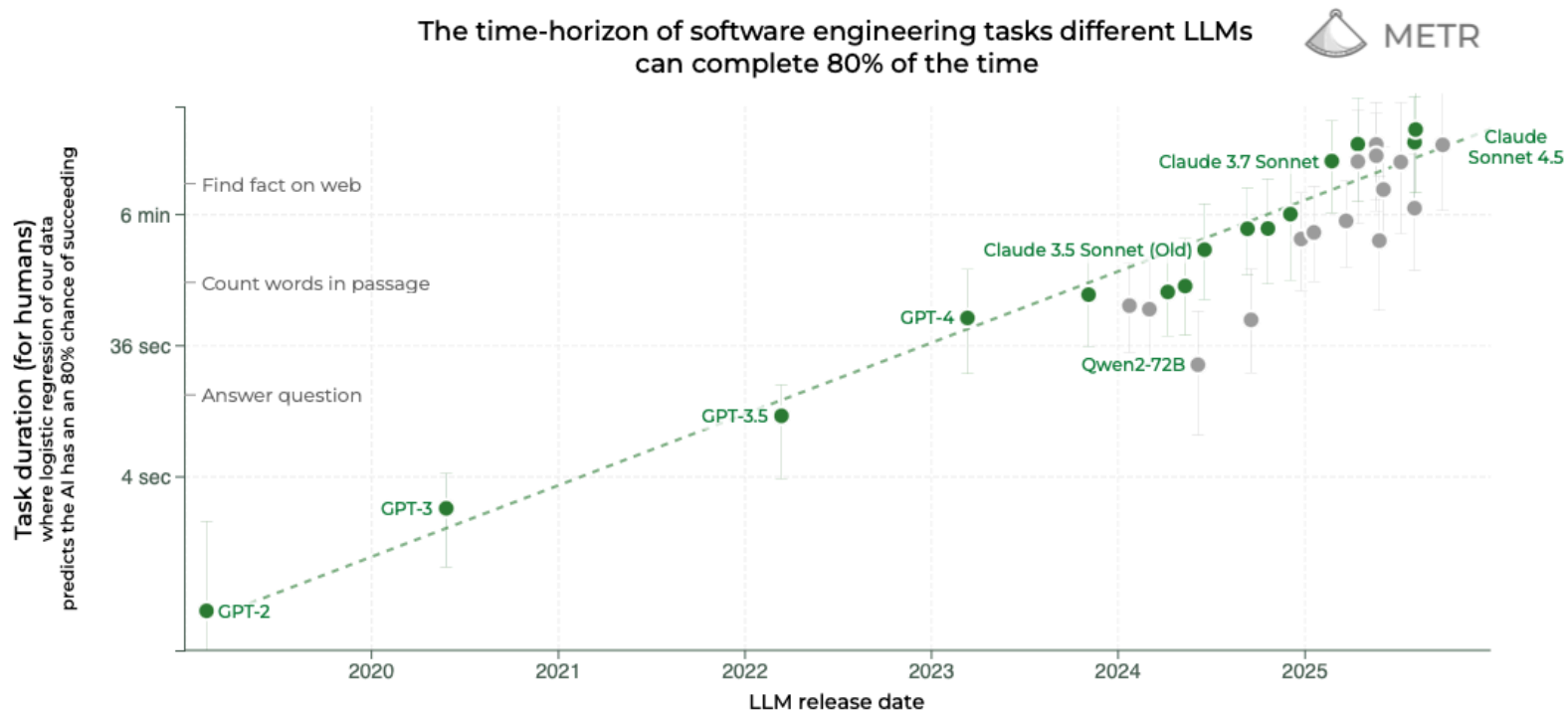
Other Approaches (Human Evals)



<https://arxiv.org/pdf/2503.14499>

<https://metr.org/blog/2025-03-19-measuring-ai-ability-to-complete-long-tasks/>

Other Approaches (Human Evals)



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Human Evals

Open source crowdsourcing evals!

Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference

Wei-Lin Chiang^{*1} Lianmin Zheng^{*1} Ying Sheng² Anastasios N. Angelopoulos¹ Tianle Li¹ Dacheng Li¹
Banghua Zhu¹ Hao Zhang³ Michael I. Jordan¹ Joseph E. Gonzalez¹ Ion Stoica¹

		Question Source	
		Static	Live
Evaluation Metric	Ground Truth	MMLU, HellaSwag, GSM-8K	Codeforces Weekly Contests
	Human Preference	MT-Bench, AlpacaEval	Chatbot Arena

Human Evals

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Best soil mix for succulents?

Assistant A

Best Succulent Soil Mix

Simple Recipe

- 60% cactus/succulent soil (commercial base)
- 40% perlite or coarse sand

This improves drainage significantly.

From Scratch

- 40% potting soil
- 40% perlite or pumice
- 20% coarse sand or grit

Key Principles

- Must drain quickly — succulents rot in wet soil
- Use pots with drainage holes
- Avoid regular potting soil — too moisture-retentive
- Add grit (not fine sand) for texture and air pockets

Shopping Shortcuts

- Pre-made "cactus mix" works fine if you add 20-30% extra perlite

Assistant B

The ideal soil mix for succulents needs to provide excellent drainage and aeration while allowing roots to breathe. Here's what works best:

Basic DIY Mix (Simple & Effective):

- 2 parts potting soil (cactus/succulent mix preferred)
- 1 part perlite or pumice
- 1 part coarse sand (horticultural grade)

Premium DIY Mix (For Better Drainage):

- 1 part potting soil
- 1 part perlite
- 1 part pumice
- 1 part coarse sand

Key Components:

- Inorganic materials (perlite, pumice, coarse sand) for drainage
- Minimal organic matter to prevent water retention
- Gritty texture to prevent compaction

What to Avoid:

- Regular garden soil (too dense)
- Peat-based mixes (retain too much water)
- Fine sand (compacts easily)

Pro Tip:

← Left is Better

It's a tie

Both are bad

Right is Better →

Human Evals

The Leaderboard Illusion

Shivalika Singh^{*1}, Yiyang Nan¹, Alex Wang², Daniel D'souza¹,
Sayash Kapoor³, Ahmet Üstün¹, Sanmi Koyejo⁴, Yuntian Deng⁵,
Shayne Longpre⁶, Noah A. Smith^{7,8}, Beyza Ermiş¹,
Marzieh Fadaee^{*1}, and Sara Hooker^{*1}

¹Cohere Labs, ²Cohere, ³Princeton University, ⁴Stanford University, ⁵University of Waterloo,
⁶Massachusetts Institute of Technology, ⁷Allen Institute for Artificial Intelligence, ⁸University of
Washington

What's In My Human Feedback? Learning Interpretable Descriptions of Preference Data

Rajiv Movva^{*1}, Smitha Milli², Sewon Min¹, and Emma Pierson¹

¹UC Berkeley

²FAIR at Meta

October 2025

<https://arxiv.org/abs/2506.13776>

<https://arxiv.org/pdf/2309.16349>

<https://arxiv.org/pdf/2504.20879>

<https://arxiv.org/pdf/2510.26202>

Human Evals

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⁶Massachusetts Institute of Technology, ⁷Allen Institute for Artificial Intelligence, ⁸University of
Washington

What's In My Human Feedback?

Learning Interpretable Descriptions of Preference Data

Rajiv Movva^{*1}, Smitha Milli², Sewon Min¹, and Emma Pierson¹

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²FAIR at Meta

October 2025

Position: Human Baselines in Model Evaluations Need Rigor and Transparency (With Recommendations & Reporting Checklist)

Kevin L. Wei^{*1,2} Patricia Paskov^{*1,3} Sunishchal Dev^{*1,4} Michael J. Byun^{*1,3} Anka Reuel^{2,5}
Xavier Roberts-Gaal² Rachel Calcott² Evie Coxon⁶ Chinmay Deshpande⁷

HUMAN FEEDBACK IS NOT GOLD STANDARD

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<https://arxiv.org/abs/2506.13776>

<https://arxiv.org/pdf/2309.16349>

<https://arxiv.org/pdf/2504.20879>

<https://arxiv.org/pdf/2510.26202>

Take-Aways

Predicting capabilities of future (and even current) AI systems is hard

(Training) compute is a flawed estimator for model capabilities

Emergent abilities pose safety concerns

Evaluations are also metric-dependent

Eliciting skills/knowledge from AI systems is an unsolved problem

Human performance baselines are not perfect

PITFALLS OF EVIDENCE-BASED AI POLICY

Stephen Casper
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David Krueger
Mila

Dylan Hadfield-Menell
MIT CSAIL

AI auditing: The Broken Bus on the Road to AI Accountability

Abeba Birhane
Mozilla Foundation and
Sch. of Computer Science and Statistics
Trinity College Dublin
Dublin, Ireland

Ryan Steed
Heinz College & Machine Learning Dept.
Carnegie Mellon University
Pittsburgh, USA

Victor Ojewale
Ctr. for Technological Responsibility,
Reimagination, & Redesign
Brown University
Providence, USA

Briana Vecchione
Data & Society
New York City, USA

Inioluwa Deborah Raji
Mozilla Foundation and
University of California, Berkeley
Berkeley, USA

Complex Systems are Hard to Control

APR 3, 2023 • 11 MIN READ

<https://arxiv.org/pdf/2502.09618>
<https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=10516659>
<https://bounded-regret.ghost.io/complex-systems-are-hard-to-control/>

Appendix

lecture content

mention freeze out and multi peak vs single peak narrowness when talking about RL?

better bench + construct validity

- issues with benchmarks (design criteria, maybe touching on construct validity a bit, maybe mention ChatBot arena) and good examples of evals (e.g., METR studies or ImpossibleBench)

I'll probably add Stephen Casper's "pitfalls of evidence-based policy" and Jacob's old post about "complex systems are hard to control" to the reading list.

mention unfaithful CoT + UQ?

METR reward hacking in coding agents

<https://metr.org/blog/2025-06-05-recent-reward-hacking/>

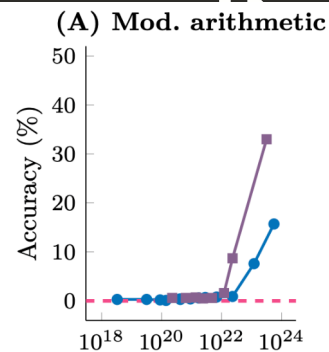
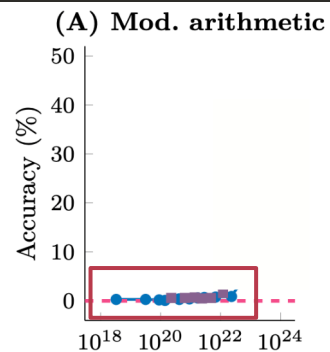
"AI auditing: The Broken Bus on the Road to AI Accountability"

<https://ieeexplore.ieee.org/abstract/document/10516659>

BetterBench https://proceedings.neurips.cc/paper_files/paper/2024/hash/26889e8359e7ef8a7f5d77457364ca55-Abstract-Datasets_and_Benchmarks_Track.html

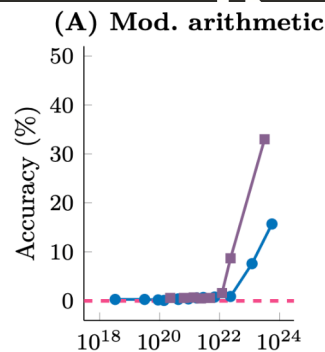
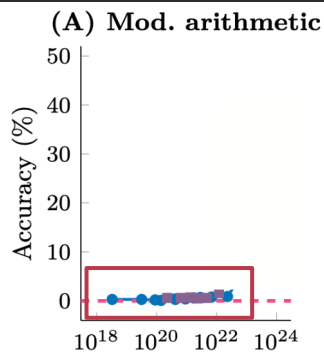
Not So Simple

“We call into question the claim that LLMs possess emergent abilities, [i.e.,] sharp and unpredictable changes in model outputs as a function of model scale on specific tasks.”



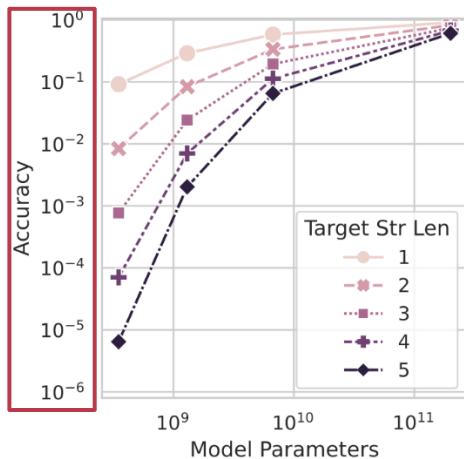
Not So Simple

“We call into question the claim that LLMs possess emergent abilities, [i.e.,] sharp and unpredictable changes in model outputs as a function of model scale on specific tasks.”

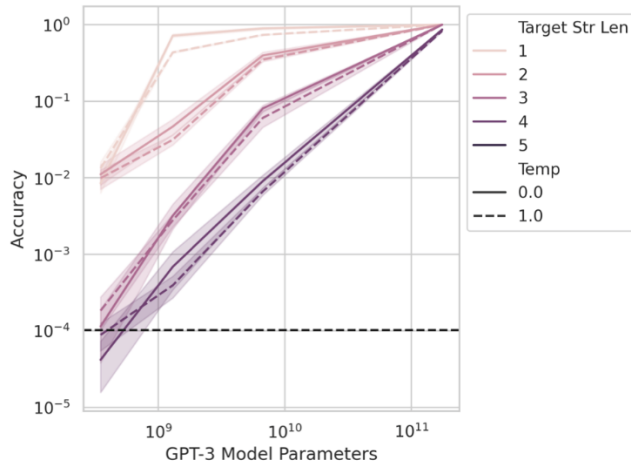


Emergence effect can diminish with more data (statistics)

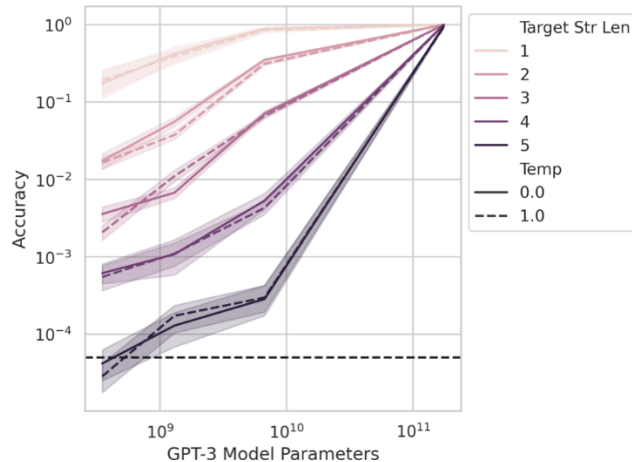
Mathematical Model



2-Integer 2-Digit Multiplication



2-Integer 4-Digit Addition





STaR-GATE: Teaching Language Models to Ask Clarifying Questions

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Self-Supervised Alignment with Mutual Information Learning to Follow Principles without Preference Labels

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Tobias Gerstenberg **Noah D. Goodman**

Stanford University



Emergent Abilities

Extensively testing multimodal AI systems is expensive and difficult

- Large action space (possible inputs)
- Hard to anticipate and test all failure modes (misinformation, non-consensual porn, copyright infringement, ...)

Failures are found *after deployment*, only get patched over time

Emergent abilities (*something you weren't necessarily testing for*) have potential for significant safety issues

So far,

Emergent abilities can be explained with improved language modeling (e.g., in-context learning) alone

No empirical evidence of other internal model reasoning

However, emergent abilities are still an issue to study

It is not just metrics



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For example, biosecurity/synthetic pathogens
(does not matter if a linear or non-linear metric is used)

“Having the ability itself is binary”

